

DONGHAE, Korea, Republic of

WMO: 471060

Lat: 37.5069N

Lon: 129.1244E

Elev: 41

StdP: 100.83

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.0	(c) -5.3	(d) -25.0	(e) 0.4	(f) -4.9	(g) -22.5	(h) 0.5	(i) -2.9	(j) 10.7	(k) -1.4	(l) 9.0	(m) -1.1	(n) 4.0	(o) 270	(p) 0.381

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 5.1	(c) 31.0	(d) 23.9	(e) 29.6	(f) 23.4	(g) 28.3	(h) 23.0	(i) 25.5	(j) 28.8	(k) 24.8	(l) 28.1	(m) 24.1	(n) 27.2	(o) 2.8	(p) 110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.6	19.6	27.5	23.8	18.8	26.9	23.1	18.0	26.2	78.9	29.1	76.0	28.2	73.1	27.3	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.1	(b) 6.8	(c) 5.9	(e) -10.1	(f) 33.7	(g) 2.4	(h) 1.5	(i) -11.8	(j) 34.8	(k) -13.2	(l) 35.6	(m) -14.5	(n) 36.5	(o) -16.3	(p) 37.5		
(a) 8.1	(b) 6.8	(c) 5.9	(e) -12.3	(f) 26.1	(g) 2.1	(h) 0.8	(i) -13.8	(j) 26.7	(k) -15.0	(l) 27.1	(m) -16.2	(n) 27.6	(o) -17.8	(p) 28.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	13.0	1.2	2.7	6.7	12.2	16.6	19.7	23.5	24.3	19.8	15.3	9.3	3.5	(6)
(7)		DBStd	8.50	3.35	3.47	3.89	3.93	3.52	2.78	3.16	2.64	2.37	2.86	3.62	3.68	(7)
(8)		HDD10.0	877	273	206	117	20	1	0	0	0	0	3	56	202	(8)
(9)		HDD18.3	2453	532	438	360	188	77	16	1	0	11	100	271	459	(9)
(10)		CDD10.0	1956	0	1	16	87	205	289	419	442	295	166	35	1	(10)
(11)		CDD18.3	491	0	0	1	4	23	56	162	183	56	5	0	0	(11)
(12)		CDH23.3	2898	0	0	1	18	107	204	1115	1284	164	5	0	0	(12)
(13)		CDH26.7	704	0	0	0	2	15	33	318	317	19	0	0	0	(13)
(14)	Wind	WSAvg	2.5	3.0	2.8	2.9	2.9	2.5	2.0	1.9	1.9	2.3	2.5	2.7	2.8	(14)
(15)	Precipitation	PrecAvg	1285	44	43	60	77	79	123	227	251	202	80	65	33	(15)
(16)		PrecMax	1819	85	128	154	179	189	376	755	760	465	304	218	161	(16)
(17)		PrecMin	847	6	4	13	13	11	19	59	57	44	21	7	5	(17)
(18)		PrecStd	294	24	33	35	41	44	78	136	145	134	62	54	33	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.8	14.3	20.8	25.8	28.6	29.6	32.9	33.1	28.8	23.8	19.9	14.6	(19)
(20)			MCWB	6.7	7.9	11.1	14.1	15.7	19.2	24.3	24.9	23.2	16.4	13.1	8.8	(20)
(21)		2%	DB	9.2	11.5	17.3	22.6	26.0	27.4	31.2	31.1	26.6	22.2	18.1	12.4	(21)
(22)			MCWB	4.0	6.0	9.3	12.9	15.4	19.3	23.8	24.6	21.5	16.4	12.3	7.3	(22)
(23)		5%	DB	7.6	9.7	15.2	20.5	24.2	25.6	29.8	29.6	25.1	21.1	16.6	10.6	(23)
(24)			MCWB	2.9	5.0	8.5	11.7	14.9	18.8	23.6	24.3	20.9	16.0	11.5	5.7	(24)
(25)		10%	DB	6.2	8.2	13.1	18.5	22.5	24.1	28.5	28.4	23.9	20.0	15.2	9.1	(25)
(26)			MCWB	2.0	3.7	7.2	11.1	14.3	18.8	23.3	24.1	20.1	15.6	10.5	4.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.3	9.2	12.4	16.7	18.6	22.3	26.2	26.6	24.1	19.5	15.6	10.3	(27)
(28)			MCDWB	10.0	12.0	18.0	22.3	23.8	25.3	29.5	29.9	27.8	21.2	17.9	13.5	(28)
(29)		2%	WB	5.1	7.1	10.6	14.4	17.3	21.4	25.3	25.8	22.6	18.4	13.9	8.1	(29)
(30)			MCDWB	7.9	10.5	15.5	20.1	22.2	24.2	28.8	29.0	25.4	20.5	16.9	11.3	(30)
(31)		5%	WB	3.8	5.6	9.2	13.3	16.5	20.7	24.6	25.2	21.7	17.5	12.5	6.5	(31)
(32)			MCDWB	6.5	8.9	13.4	18.2	21.1	23.4	28.2	28.3	24.1	19.9	15.4	9.7	(32)
(33)		10%	WB	2.6	4.2	7.9	12.2	15.7	20.0	23.8	24.6	20.8	16.4	11.2	5.2	(33)
(34)			MCDWB	5.5	7.4	12.3	16.6	20.1	22.6	27.4	27.6	23.2	19.1	14.4	8.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.8	6.9	7.2	7.5	6.9	5.3	4.7	5.1	6.1	7.4	7.3	7.1	(35)
(36)			MCDBR	8.0	8.6	9.6	10.6	9.8	7.8	7.1	6.8	7.2	8.5	8.5	8.4	(36)
(37)			MCWBR	5.7	6.2	6.0	5.8	4.8	3.4	3.0	3.2	4.6	5.4	6.1	6.2	(37)
(38)		5% WB	MCDBR	7.0	8.0	8.6	8.7	7.9	5.5	5.7	5.9	6.4	6.8	7.1	7.7	(38)
(39)			MCWBR	5.7	6.5	6.2	5.7	4.7	3.1	3.2	3.3	4.4	4.9	5.7	6.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.416	0.477	0.571	0.608	0.614	0.661	0.623	0.560	0.489	0.478	0.444	0.407	(40)	
(41)		taud	1.992	1.872	1.708	1.687	1.719	1.685	1.830	1.990	2.119	2.015	2.029	2.045	(41)	
(42)		Ebn at Noon	721	723	693	700	706	673	695	731	762	717	687	699	(42)	
(43)		Edh at Noon	132	170	220	237	234	242	208	173	144	145	127	118	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.37	3.14	4.08	4.87	5.40	5.12	4.31	4.26	3.85	3.37	2.42	2.15	(44)	
(45)		RadStd	0.20	0.31	0.40	0.37	0.52	0.44	0.61	0.53	0.46	0.30	0.31	0.18	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	-1.23	N/A	N/A	N/A	N/A	N/A	+120	+55	(46)
(47)	Regional (35 neighbors)	N/A	N/A	-1.00	N/A	N/A	N/A	N/A	N/A	+93	+50	(47)

Nomenclature: See separate page